

Global Measurement and Geospatial Analysis of the I2P Anonymous Communication Network

Cavel Campbell, INCS Project I

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All figures, tables, and statistical tests in this report are computed from routerInfo observations collected July 7–17, 2026 (see §4.1), as captured in data/i2p.duckdb.

Abstract

The Invisible Internet Project (I2P) is one of the most widely used anonymity networks after Tor, but it has not received the same level of academic attention. This study aims to help close that gap by carrying out a focused measurement analysis of the I2P network. Using two independent vantage points, a Java I2P router and an i2pd router, we collected routerInfo records describing 48,235 distinct routers across 161 countries and 2,866 Autonomous Systems.

The results show that I2P adoption is uneven across the world and that different regions have noticeably different structural patterns. The United States and Iran contain the two largest national router populations, with 6,376 and 2,471 routers respectively. DBSCAN clustering over router coordinates reveals a small number of dense geographic clusters. One of the largest clusters, containing 2,442 routers, is centered almost directly on Tehran and is served mostly by two Iranian telecommunications ASNs that are tied to state infrastructure. This is different from the diverse ISP mix seen in Western regions.

One of the original research questions asked whether it was possible to identify which I2P client implementation each router used. This cannot be answered through passive netDb observation. The router.version field only reports the shared protocol version and does not reveal the underlying implementation. This is reported as a methodological finding rather than an unmet goal.

Statistical testing shows that geography and network behavior are meaningfully connected. Protocol version ($\chi^2=4416.1$, Cramér's $V=0.17$), floodfill adoption ($\chi^2=1867.4$, $V=0.34$), and bandwidth tier ($\chi^2=5923.8$, $V=0.35$) are all significantly associated with country even after Bonferroni correction. Geographic DBSCAN clusters also differ in router level risk score (Kruskal Wallis $H=2174.1$, $p<0.001$, $\epsilon^2=0.090$). Out of eight behavioral metrics computed, only one pair, risk score and suspiciousness ($r=0.964$), is redundant enough to represent the same underlying signal. The remaining metrics capture distinct information about router behavior.

1. Introduction

Anonymity networks are often studied as tools for privacy and protection, and sometimes as platforms for illicit activity, but the attention they receive is not evenly distributed. Tor has been examined in depth for years. Researchers have analyzed its consensus system, relay churn, exit node geography, and overall performance. I2P, even though it is similarly mature and built on a very different design, has not received the same level of empirical study. I2P uses garlic routing, a decentralized network database called the netDb, and a peer to peer discovery model. It does not rely on centralized directory authorities the way Tor does. Because of this, many basic questions about I2P remain open. We still do not have a clear picture of how the network is deployed globally, how routers differ in capability and version currency, or how users are distributed geographically.

This report addresses that gap through a six to eight week measurement study that combines a network census with geospatial analysis. The work is guided by three main research questions.

1. Version and capability distribution. What protocol versions, capability flags, and bandwidth classes appear across the network, and do outdated or under capable routers cluster in specific regions?
2. Geographic structure. How are I2P routers distributed around the world, and do they form identifiable geographic clusters instead of a uniform global spread?
3. Correlation between geography and behavior. Do geographic patterns line up with differences in version currency, capability mix, or uptime? In other words, do different regions show distinct usage patterns, not just different population sizes?

A fourth question was part of the original scope, but it turned out to be unanswerable with passive observation. The goal was to determine which I2P client implementation (Java I2P, i2pd, or I2P+) is most common across the network. Sections 6.1 and 7.1 explain why this cannot be determined from netDb data and what can be measured instead, such as protocol version distribution and staleness.

The rest of this report is organized as follows. Section 2 provides background on I2P's architecture. Section 3 places this work in the context of prior Tor and I2P measurement research. Section 4 describes the methodology, including data collection, enrichment, and analysis, and explains where the completed work matched, exceeded, or diverged from the original project proposal. Section 5 describes the implementation of the system used to conduct the study. Sections 6 and 7 present the results. Section 8 discusses their interpretation. Section 9 covers limitations. Section 10 outlines future work. Section 11 concludes the report.

2. Background

I2P is a fully decentralized, peer to peer anonymity network. Every participant runs a router, and routers exchange signed records called routerInfo entries. These records describe the router's network addresses, transport protocols, capability flags, bandwidth class, and publish timestamp. Routers share these records through gossip, which builds the netDb, a distributed and eventually consistent view of the network. Unlike Tor, I2P does not rely on a small group of trusted directory authorities to publish a complete network state. Each router only sees what it has learned through its own gossip interactions. This design has a direct impact on how the network can be measured. A single observation point only sees a partial slice of the network, shaped by whichever peers it happens to gossip with. Because of this, using two independent vantage points is important. In this study, one vantage point is a Java I2P router and the other is an i2pd router, which is a separate C++ implementation. Using both allows gaps in one router's view to be compared against the other instead of being mistaken for true absence.

I2P routers advertise several properties that matter for this study.

Capability flags. These encode floodfill status, congestion state, reachability, and bandwidth tier.

Bandwidth class. This is a coarse tier that indicates the router's self reported bandwidth level. The tiers roughly map to low, mid, high, and unlimited.

Protocol version. This is a version string such as 0.9.68 that reflects the I2NP protocol version the router speaks. It is important to note that this is a protocol version shared across client implementations. It does not identify which software produced the routerInfo entry, which becomes a key methodological point later in the report.

3. Related Work

Research on anonymity networks has grown steadily, but most of that attention has focused on Tor rather than I2P. Tor measurement is a mature area. McCoy et al. (2008) established one of the foundational approaches by running a modified Tor relay to study live traffic instead of relying on simulations. This report's router-based method for I2P follows the same general idea. Loesing, Murdoch, and Dingledine (2010) expanded Tor measurement by introducing systematic and privacy-preserving data collection from relays and directory authorities, which helped shape the Tor Project's long-running Metrics effort. Winter, Ensafi, Loesing, and Feamster (2016) analyzed nine years of archived Tor data to identify Sybil relay groups using configuration and behavioral fingerprints. Their work is methodologically close to this study's use of DBSCAN clustering and chi square testing, since both approaches rely on structural metadata to uncover patterns the network does not label on its own. Across the Tor literature, geographic and infrastructural concentration is a recurring theme for understanding how a decentralized network behaves in practice. This same lens is applied to I2P in Section 7.

I2P-specific measurement research exists, but it is far more limited. Timpanaro, Chrisment, and Fester (2012) conducted the first published monitoring study of I2P, focusing on applications visible from a single vantage point over one-week windows. This report's ten-day, two-vantage-point census extends that scale directly. Egger, Schlumberger, Kruegel, and Vigna (2013) demonstrated practical deanonymization attacks against I2P hidden services. Their work highlights why this study relies strictly on passive, aggregate-level measurement rather than any technique that could compromise user anonymity. Hoang, Antonakakis, Kintis, and Polychronakis (2018) carried out the largest prior empirical study of I2P, measuring roughly 32,000 daily active peers, churn, router types, geographic distribution, and censorship resistance. Their scope is the closest to this report, although their observed population is about two-thirds the size of the dataset used here and does not include formal statistical testing. Liu, Zhang, Shi, Yu, and Xu (2019) analyzed about 16,000 I2P nodes for country, bandwidth, and floodfill attributes, which aligns directly with the capability and geographic results presented in Sections 6.3 and 7.1.

One study is especially relevant to this report's findings about Iran. Hoang, Doreen, and Polychronakis (2019) documented active state-level I2P blocking in five countries, including Iran. Their work provides independent evidence that Iran is a known site of I2P censorship. This context helps interpret the structural patterns observed in this report, such as Iran's concentrated router presence and its distinct ASN distribution. However, neither their study nor this one can determine individual user intent from

network-level data. Both rely on aggregate observation rather than any form of traffic correlation or deanonymization.

4. Methodology

4.1 Data Collection

This study uses two independent I2P router processes as vantage points: a Java I2P router and an i2pd router. Each maintains its own local netDb. Every observation is tagged with the vantage point that produced it, which allows direct comparison instead of merging the data blindly. This design supports the goal of understanding how much of the network each implementation can see and where their views differ.

A collection script walks both netDb directory trees and extracts each router's identity hash, IP address, transport protocols, capability flags, version string, and publish timestamp. More detail on how these fields are parsed appears in Section 5.2.

Collection ran on a fixed 30 minute cycle from July 7 to July 17, 2026. This ten day window matches the minimum target in the project proposal. A single snapshot would not be enough to measure uptime, churn, or observation regularity, and it would not distinguish a genuinely small network from a large one that happened to be partially unseen at one moment. Running the pipeline continuously solves these problems.

During development, several data quality issues were found and corrected. A decoding bug caused boolean fields from the Java router's wire format to be corrupted, since every boolean was stringified to a truthy value. A deduplication bug also inflated the raw observation table until a primary key correction reduced the row count from 452,156 to 178,049. Both issues were fixed before analysis began. These corrections are part of the method and not hidden engineering cleanup.

4.2 Data Enrichment

Each router's IP address was resolved to geographic coordinates, country, and Autonomous System using MaxMind's GeoLite2 City and ASN databases. An early version of this step only attempted IPv4 parsing, which would have excluded IPv6 only routers. Roughly 18 percent of routers were IPv6 only at the time this was caught. MaxMind resolves IPv6 with similar accuracy to IPv4, so the enrichment step was corrected to use whichever address family the router advertised. Without this fix, IPv6 heavy regions would have been underrepresented in every geographic result.

Capability flag parsing was validated against the published netDb specification rather than relying on assumptions. This surfaced two issues. One capability letter reflected congestion state instead of introducer support, and another reflected a bandwidth tier instead of peer test support. Bandwidth tier bucketing also needed correction to cover the full range of advertised tiers.

4.3 Analysis Methods

Geographic clustering.

Router coordinates were clustered using DBSCAN with a haversine distance metric. The primary radius was 250 kilometers with a minimum cluster size of five routers. Routers outside any sufficiently dense group were labeled as noise instead of being forced into a cluster. Because the radius is a methodological choice, not a ground truth, a sensitivity analysis repeated the clustering at 100 kilometers and 500 kilometers. Cluster assignments were compared back to the 250 kilometer baseline using the Adjusted

Rand Index, which measures how consistently routers end up grouped together across different radius choices. You can revisit this concept through the Adjusted Rand Index.

Statistical association between geography and behavior

Three chi square tests of independence were used to test whether protocol version, floodfill capability, and bandwidth tier are distributed independently of the country. These tests were restricted to the ten countries with at least twenty observed routers to keep contingency table cells adequately populated. Because three tests were run on the same country groupings, a Bonferroni correction was applied to control the family wise error rate. Cramér's V with Bergsma's bias correction was computed as an effect size measure, since large sample sizes can make nearly any association statistically significant. You can revisit this through Cramér's V.

Wilson score confidence intervals were used for per-country proportions such as floodfill and IPv6 adoption. Several countries in the dataset have as few as five routers, and Wilson intervals perform better than normal approximation intervals in small samples.

A Kruskal Wallis H test was used to evaluate whether DBSCAN geographic clusters differ in risk or anomaly score distributions. These distributions are right-skewed, so a non-parametric test is appropriate. Effect size was measured with epsilon squared, which is similar to eta squared in ANOVA.

Finally, Pearson correlation was computed across all pairs of eight router-level metrics: risk score, anomaly score, suspiciousness, entropy, periodicity, stability, cluster distance, and uptime. Any pair with an absolute correlation of at least 0.7 was flagged as redundant. This identifies metrics that represent the same underlying signal instead of independent evidence. You can revisit this through Pearson correlation.

4.4 Ethical Considerations

This study relies exclusively on passive observation of publicly gossiped netDb records: the same information any I2P participant necessarily receives simply by participating in the network. No attempt was made to deanonymize individual users, correlate traffic content, or otherwise compromise the anonymity properties I2P is designed to provide. Raw IP addresses are excluded from any exported dataset; the analysis dataset retains only derived fields (country, ASN, capability flags, coordinates rounded to city level precision) sufficient for the aggregate, population level questions this study asks.

4.5 Adherence to and Deviations from the Proposed Scope

The project proposal defined a narrow scope: two vantage point routers, a seven to ten day window, MaxMind-based enrichment, statistical evaluation using DBSCAN or K-means, and charts, maps, and written analysis. It explicitly excluded multi-month studies, deep machine learning models, and complex dashboards. This section explains where the completed work matched, exceeded, or diverged from that scope.

Matched or exceeded

Collection used the proposed two vantage points and ran the full ten day window with continuous automation instead of manual snapshots. This strengthened metrics such as uptime, churn, and periodicity, which require multiple observations per router. Enrichment used the proposed MaxMind pipeline plus additional correctness work, including the IPv6 fix and capability flag corrections. Geographic clustering used DBSCAN, which was one of the proposed options, and was chosen because it does not require a fixed number of clusters and separates noise from dense regions. Statistical analysis went beyond the proposal's general call for evaluation by adding formal significance and effect size testing.

Built beyond the proposed scope

A larger analytical system was built alongside the main pipeline. This includes an IsolationForest anomaly detector, PCA-based ASN reputation scoring, K-means behavioral clustering, a fifteen-component router risk scoring engine, and a Flask dashboard. These tools go beyond the proposal's instructions to avoid deep machine learning and complex dashboards. None of their output appears in this report, since they do not answer the study's three research questions. They are disclosed here and mentioned again in Section 10 as possible directions for future work.

Not answerable as proposed

The proposal's central question about identifying which client implementation is most prevalent cannot be answered with passive netDb data. The netDb does not expose an implementation identifier. This is explained in Sections 6.1, 7.1, and 8. Protocol version distribution and staleness can be measured instead. The inability to identify implementations is reported as a methodological finding rather than a failure.

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5. Implementation

Section 4 describes the methodological design of the study. This section explains how that design was built in practice, including the running system, its components, and the reasoning behind each technical choice.

5.1 Pipeline Architecture

The system operates as a nine-stage Python pipeline that runs end to end on every 30 minute collection cycle. The stages are:

1. raw netDb ingestion
2. timeseries and churn computation
3. ASN mapping and enrichment
4. ASN analytics
5. machine learning engines
6. risk scoring engines
7. geographic history and clustering
8. behavioral metric rebuilding
9. visualization

Only stages 1 through 3, 7, and 9 contribute to the results in this report. Stages 4 through 6 implement the anomaly detection, ASN reputation scoring, and risk scoring components described earlier in Section 4.5. These components run as part of the pipeline but do not influence any figure, table, or statistical result presented here. You can revisit these ideas through pipeline design or anomaly detection.

5.2 Router Instrumentation and netDb Parsing

The two vantage point routers, Java I2P 2.12.0 and i2pd 2.60.0, run as persistent background services under macOS launchd. A supervisor script triggers a full collection and pipeline cycle every 30 minutes.

i2pd's routerInfo files are parsed using a custom Python decoder that handles I2P's binary and signed format. When a record does not decode cleanly, the parser falls back to text scanning. Java I2P's netDb is parsed differently. Instead of re-implementing the format, the pipeline loads the router's own Java classes through JPy and reads the records directly. This approach is why the boolean corruption bug described in Section 4.1 was caught on the Java side rather than the i2pd side.

Implementation identity is handled through a small lookup table that maps the two self-controlled routers to hardcoded labels. Every other router's implementation field is left NULL by design. This reflects the methodological finding in Section 6.1 that implementation identification is not possible through passive netDb observation. You can revisit this through I2P netDb or routerInfo parsing.

5.3 Storage and Enrichment

Parsed observations are stored in a single DuckDB file. DuckDB was chosen because it supports fast columnar aggregation without requiring a separate database server. The dataset contains 789,283 raw location observations and 48,235 deduplicated routers.

Geographic and ASN enrichment uses MaxMind's GeoLite2 City and ASN databases through the geoip2 library. The enrichment step resolves whichever address family the router advertises, which ensures IPv6-only routers are included. The output table, `enriched_router_data`, is the canonical dataset used for all statistics, tables, and figures in this report. You can revisit this through DuckDB or MaxMind geolocation.

5.4 Statistical and Clustering Implementation

Geographic clustering uses DBSCAN from scikit-learn as its own pipeline stage. It operates only on router latitude and longitude and does not share code with the K-means clustering used in the machine learning components described in Section 4.5. This separation ensures that the geographic results in Section 7 are independent of the excluded ML machinery.

The chi square tests, Kruskal Wallis tests, Mann Whitney U tests, and Wilson score intervals described in Section 7.6 run directly against `enriched_router_data` and its join tables. All eight figures in the report were generated with matplotlib from these same tables. Every number in the text and every point or bar in the charts traces back to this shared source. You can revisit these concepts through DBSCAN or Kruskal_Wallis.

6. Results: Version, Capability, and Bandwidth Distribution

6.1 The implementation identification finding

One of the original research questions asked whether it was possible to identify which I2P client implementation each router used. This turned out to be unanswerable with passive netDb data. The `router.version` field only reports the I2NP protocol version, such as 0.9.68, and this version is shared across different client implementations. It does not reveal whether the routerInfo entry came from Java I2P, i2pd, or I2P+.

This was confirmed directly. The Java I2P router used in this study (version 2.12.0) and the i2pd router (version 2.60.0) both reported the same protocol version string, 0.9.69, in their routerInfo records. Since the netDb does not expose any implementation-specific field, the only routers whose implementation can be stated with certainty are the two vantage point routers controlled in this study. Every other router's implementation remains unknown by design. You can revisit this concept through I2P protocol version.

This is reported as a substantive methodological finding. It shows that passive netDb observation cannot be used for implementation fingerprinting. Any future work that wants to classify routers by implementation will need active probing or a different data source.

6.2 Protocol version distribution

Out of 48,235 observed routers, protocol version 0.9.68 was the most common, appearing on 20,352 routers (42.2 percent). The next most common versions were 0.9.69 with 9,358 routers (19.4 percent) and 0.9.64 with 7,206 routers (14.9 percent).

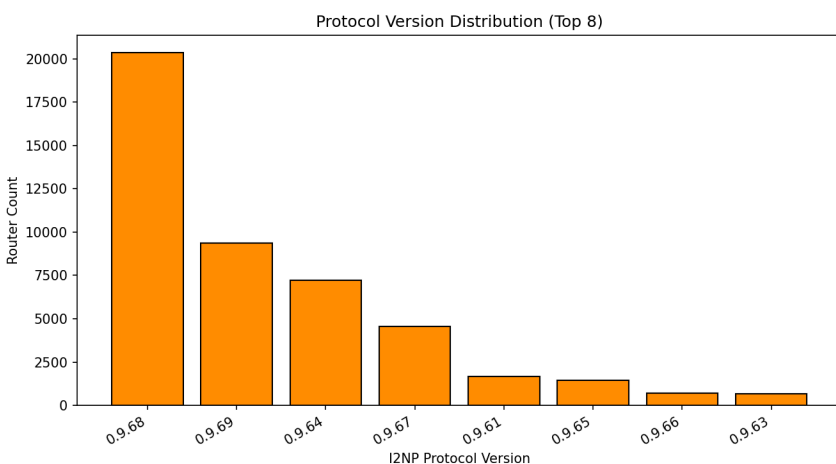


Figure 1: Protocol version distribution across 48,235 observed I2P routers, top 8 versions by count.

The version spread is concentrated in the two or three most recent releases, with a long tail of older versions making up the rest. Publish timestamp staleness, measured as the number of days since a router last republished its routerInfo, averaged 5.3 days across the network. The median was 4.8 days. Among the four largest national populations, there were no major outliers. The United States averaged 5.1 days, Iran 5.4, Russia 4.6, and Germany 3.9.

The chi square test in Section 7.6 refines this picture. Across the full version distribution and the ten major populations tested, protocol version is significantly associated with the country. The effect size, measured by Cramér's V at 0.171, is modest. This means the association is real but not as strong as the associations found for floodfill capability or bandwidth tier. You can revisit this through version distribution.

6.3 Capability and bandwidth distribution

11,626 of 48,235 routers (24.1%) advertise floodfill capability. Bandwidth tier distribution skews toward the low and high ends rather than the middle: 19,992 routers (41.4%) report the low tier, 1,249 (2.6%) the mid tier, 14,050 (29.1%) high, and 12,942 (26.8%) unlimited.

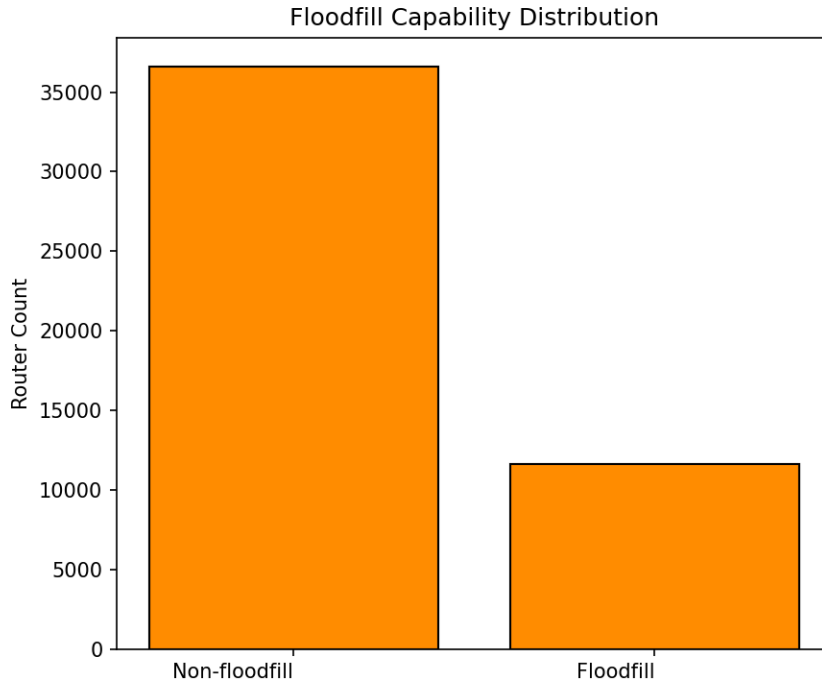


Figure 2: Floodfill capability distribution across all observed routers.

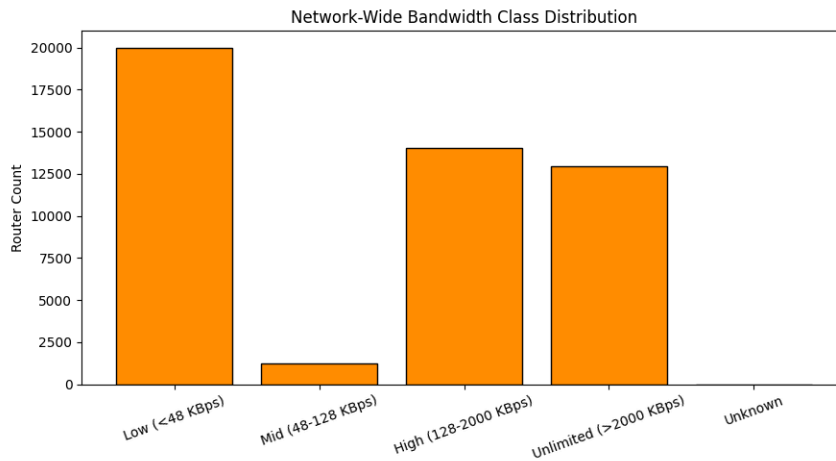


Figure 3: Network wide bandwidth class distribution.

6.4 Protocol version currency by country

A recurring question during scoping was whether different countries favor different I2P client software. Section 6.1 shows that this cannot be answered directly. What can be measured instead is protocol version currency. This serves as a reasonable proxy for update behavior.

Table 1 shows each of the eight largest populations' top three protocol versions:

Country	1st (share)	2nd (share)	3rd (share)
RU	0.9.69 (38.5%)	0.9.68 (35.2%)	0.9.67 (14.1%)

Country	1st (share)	2nd (share)	3rd (share)
GB	0.9.69 (35.8%)	0.9.64 (26.6%)	0.9.68 (19.9%)
NL	0.9.69 (33.7%)	0.9.68 (28.6%)	0.9.64 (18.8%)
CA	0.9.69 (33.7%)	0.9.64 (22.7%)	0.9.68 (22.4%)
AU	0.9.69 (30.2%)	0.9.68 (27.0%)	0.9.64 (23.9%)
DE	0.9.68 (41.7%)	0.9.69 (34.8%)	0.9.65 (7.1%)
US	0.9.64 (30.9%)	0.9.68 (25.4%)	0.9.69 (21.1%)
IR	0.9.68 (49.6%)	0.9.67 (21.7%)	0.9.69 (17.4%)

Version 0.9.69 was the dominant current release during the collection window. A small number of routers had already moved to 0.9.70, and these were grouped with 0.9.69 as part of the same current release family.

Country	n	% on current release
RU	1,741	38.5%
GB	839	35.8%
DE	1,184	34.8%
CA	843	33.8%
NL	612	33.7%
FR	597	31.7%
AU	708	30.2%
US	6,376	21.1%
IR	2,471	17.4%
BR	519	13.9%

Collapsing the data into a single currency metric, defined as the share of routers on the 0.9.69 or 0.9.70 release family, produces a clearer ranking. Russia, the United Kingdom, Germany, Canada, the Netherlands, France, and Australia all fall between 30 and 38 percent. The United States and Iran fall behind at 21.1 percent and 17.4 percent. Brazil is lowest at 13.9 percent.

This aligns with the chi square result in Section 7.6. Protocol version is significantly associated with country, but the effect size is smaller than the associations found for floodfill capability or bandwidth tier. The key point is that the largest national populations are not the most current ones. Router count and update currency are independent dimensions. You can revisit this through version currency.

7. Results: Geographic Structure and Correlation

7.1 Global distribution

The dataset contains routers from 161 countries and 2,866 Autonomous Systems. The United States has the largest number of observed routers with 6,376, which is 13.2 percent of the dataset. Iran follows with 2,471 routers, then Russia with 1,741, Germany with 1,184, and Canada with 843.

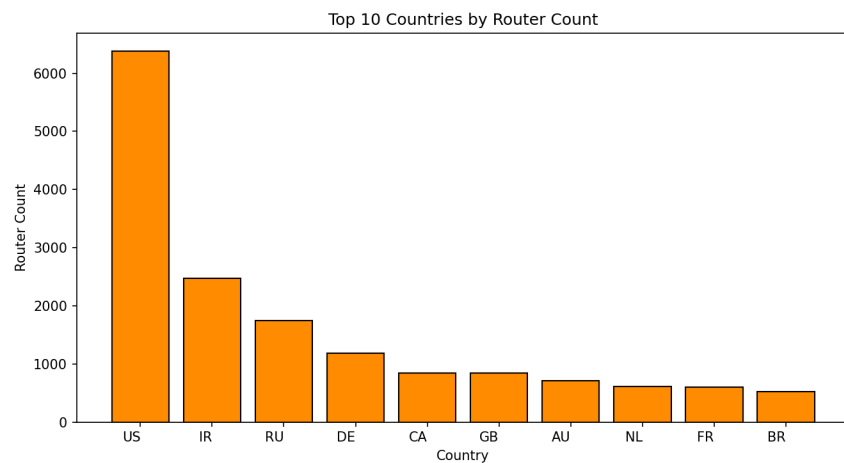


Figure 4: Top 10 countries by observed router count.

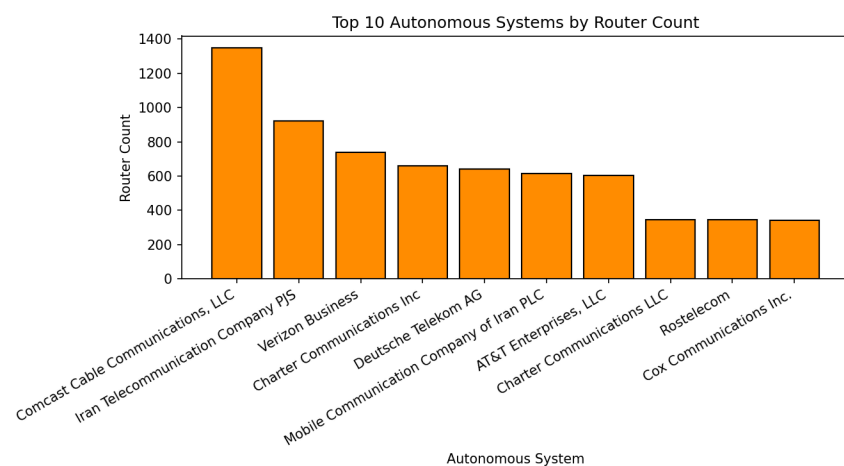


Figure 5: Top 10 Autonomous Systems by observed router count.

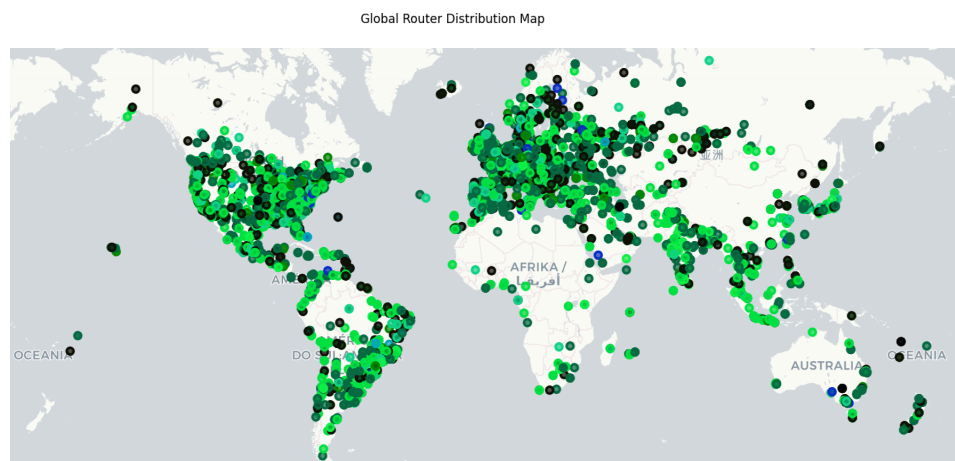


Figure 6: Global distribution of observed I2P routers.

Iran’s position as the second largest national population is notable and is examined more closely in Sections 7.3 and 8. You can revisit this through I2P geography.

7.2 Geographic clustering

DBSCAN clustering was applied to the 23,253 routers with valid coordinates. Using a 250 kilometer radius and a minimum cluster size of five, the algorithm identified 118 geographic clusters. A total of 212 routers, or 0.9 percent, were labeled as noise because they were too isolated to fall into any cluster. The three largest clusters account for a disproportionate share of clustered routers:

Cluster	Size	Approx. center
0	8,078	50.2°N, 15.6°E (Central Europe)
2	5,758	38.2°N, 86.7°W (Eastern US)
4	2,442	35.7°N, 51.4°E (Tehran, Iran)

Cluster 4’s center coordinate corresponds almost exactly to Tehran, indicating this is not a diffuse “Iran region” cluster but a genuinely dense, city centered concentration, distinct in character from the two larger clusters, which span broad multi country regions (Central Europe; the Eastern US) rather than a single metropolitan area.

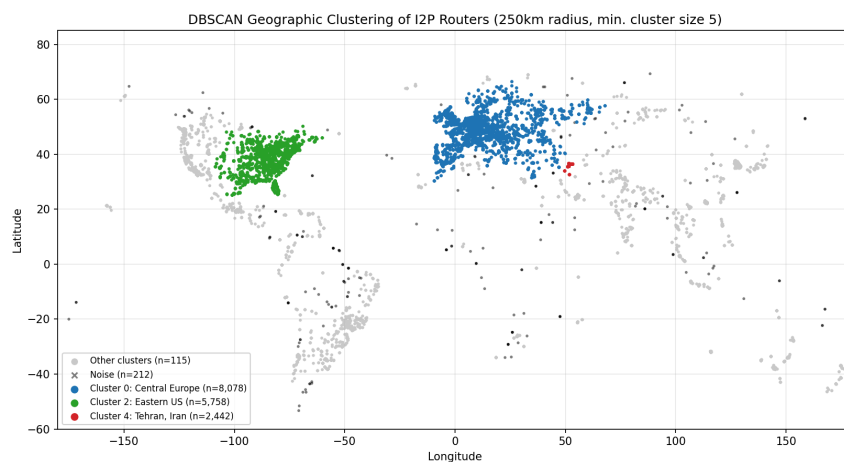


Figure 7: DBSCAN geographic clusters of observed I2P routers (250km radius, minimum cluster size 5), with the three largest clusters highlighted.

Sensitivity to clustering radius. DBSCAN was re-run at 100 kilometers and 500 kilometers to test how sensitive the results are to the chosen radius. The Adjusted Rand Index was used to compare cluster assignments back to the 250 kilometer baseline. The ARI was 0.484 at 100 kilometers and 0.647 at 500 kilometers. An ARI of 1.0 would indicate identical assignments.

Cluster and noise counts also changed. At 100 kilometers, there were 319 clusters and 3.75 percent noise. At 500 kilometers, there were 37 clusters and 0.25 percent noise.

This shows that the broad structure of the network, meaning a few large dense regions and many smaller ones, is stable. However, the exact number and boundaries of clusters depend on the radius. You can revisit this through DBSCAN radius sensitivity.

7.3 A structural difference in Iran's network presence

Iran's router population differs from other large countries in several ways. Router concentration by Autonomous System shows that countries like the United States and Germany spread their routers across many independent ISPs. No single US ASN accounts for more than about one fifth of the country's routers.

Iran's routers are concentrated in a small number of ASNs tied to state telecommunications infrastructure. Iran Telecommunication Company PJS has 820 routers, and Mobile Communication Company of Iran PLC has 503. Together, they account for more than half of Iran's geolocated routers. The next largest Iranian ASN has only 58 routers.

This combination of a large national population, a dense cluster centered on the capital, and concentration in state-linked ASNs is interpreted in Section 8. The interpretation is consistent with circumvention-driven usage over state infrastructure, although this cannot be proven from router data alone. You can revisit this through Iran I2P structure.

7.4 Network activity trend near Tehran

The geographic clustering and ASN concentration raise a follow-up question. Did activity in the Tehran cluster change over the collection window? The router_location_history table stores timestamps for every position observation, which makes this testable.

Activity was measured as the share of daily global router activity that occurred within a bounding box around Tehran. This controls for a confound in the dataset. The first full day of collection shows elevated router counts everywhere, not just in Iran. This is a global artifact caused by a freshly started router backfilling its netDb. Normalizing by the global daily total removes this artifact.

Date (2026)	Tehran area routers	Global distinct routers	Tehran share
Jul 8	554	10,557	5.25%
Jul 9	436	9,626	4.53%
Jul 10	449	8,791	5.11%
Jul 11	337	8,043	4.19%
Jul 12	439	7,710	5.69%
Jul 13	613	8,600	7.13%
Jul 14	638	9,272	6.88%
Jul 15	458	7,210	6.35%
Jul 16	619	8,858	6.99%
Jul 17	466	6,882	6.77%

Comparing the first five full days to the next five using a Mann Whitney U test shows a complete separation. Every value in the later group is higher than every value in the earlier group. The test statistic is $U = 0$ with $p = 0.0079$. This indicates a real shift in Tehran's share of network activity.

The shift is a step change rather than a gradual increase. It occurs between July 12 and July 13 and remains elevated for the rest of the window.

On timing. The timing matters because the collection window begins almost exactly when the regional conflict escalated. Iranian forces attacked commercial shipping near the Strait of Hormuz on July 7 and 8.

The IRGC struck US targets in Bahrain and Kuwait. The United States responded with counterstrikes on July 9, 10, and 11. Data collection began on July 7 at 22:32 UTC, after the escalation had already started.

This dataset cannot show any pre-event baseline. It also does not show activity rising during the most intense strike days. Tehran’s share is flat or declining on July 9, 10, and 11, and reaches its lowest point on July 11. The increase begins the next day.

A plausible interpretation is that the conflict may have prompted increased circumvention activity afterward. However, this cannot be confirmed. The dataset is short, covers only one event, and has no baseline. Other explanations remain possible, such as ISP reclassification or normal week-to-week variance. You can revisit this through Tehran activity trend.

7.5 Vantage point coverage comparison

On the last full day of the collection window, July 17, the Java I2P router observed 6,171 distinct routers. The i2pd router observed 7,626. They shared 3,262 routers in common. This is a Jaccard overlap of 31.0 percent. This relatively low overlap shows that a single vantage point would significantly undercount the network. The two implementations are not redundant. They see different parts of the network, which supports the two-vantage-point design.

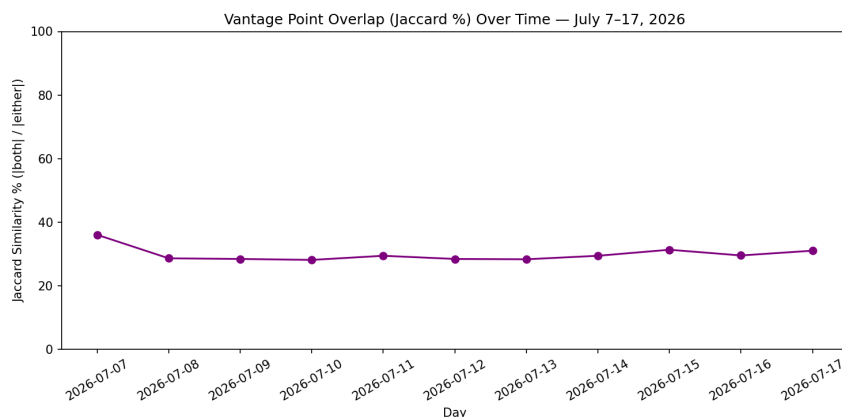


Figure 8: Daily Jaccard similarity between the Java I2P and i2pd vantage points, July 7–17, 2026.

Figure 8 shows the daily Jaccard similarity from July 7 to July 17. After a brief spike on the partial first day, overlap settles into a narrow band between 28.1 and 31.3 percent.

7.6 Geographic correlation with version, capability, and behavior

Per country proportions. Table 2 shows floodfill and IPv6 adoption with 95% Wilson confidence intervals for the ten largest national populations.

Country	n	Floodfill % (95% CI)	IPv6 % (95% CI)	Avg. uptime (h)
US	6,376	48.9% (47.7–50.2)	28.1% (27.0–29.3)	19.3
IR	2,471	8.4% (7.4–9.6)	39.0% (37.1–40.9)	18.1
RU	1,741	14.5% (12.9–16.2)	23.5% (21.6–25.6)	19.6

Country	n	Floodfill % (95% CI)	IPv6 % (95% CI)	Avg. uptime (h)
DE	1,184	22.0% (19.8–24.5)	19.4% (17.3–21.8)	28.1
CA	843	41.0% (37.8–44.4)	10.9% (9.0–13.2)	21.0
GB	839	42.3% (39.0–45.7)	9.7% (7.8–11.8)	17.0
AU	708	33.8% (30.4–37.3)	25.4% (22.4–28.8)	18.5
NL	612	35.1% (31.5–39.0)	24.3% (21.1–27.9)	20.9
FR	597	31.0% (27.4–34.8)	24.1% (20.9–27.7)	22.4
BR	519	55.5% (51.2–59.7)	44.5% (40.3–48.8)	18.7

Two gaps stand out as large enough that their confidence intervals do not overlap, meaning the difference is unlikely to be sampling noise: Iran’s floodfill rate (8.4%) is roughly a sixth of the US rate (48.9%) and non overlapping with every other country in the table except Russia, while Iran’s and Brazil’s IPv6 adoption (39.0%, 44.5%) is two to four times several Western countries’ rates (e.g., GB at 9.7%, CA at 10.9%).

Chi square tests of independence (top 10 countries by population, Bonferroni $\alpha = 0.0167$):

Test	χ^2	dof	p	Cramér’s V	Effect size	Significant (Bonferroni)
Protocol version × country	4416.1	216	<0.001	0.171	small	Yes
Floodfill × country	1867.4	9	<0.001	0.342	moderate	Yes
Bandwidth tier × country	5923.8	27	<0.001	0.352	moderate	Yes

All three associations are statistically significant even under the stricter Bonferroni threshold, and, unlike a bare p value at this sample size, the Cramér’s V effect sizes indicate these are not merely statistically detectable but practically meaningful: floodfill and bandwidth tier distribution by country reach the “moderate” effect size band, while protocol version’s association with country is real but comparatively small. This directly answers the proposal’s original question of whether outdated or distinctively configured routers cluster by region: **capability and bandwidth patterns do vary meaningfully by country; version currency varies too, but the association is weaker.** This refines the descriptive staleness observation in Section 6.2 (which found no strong staleness gap among the four largest populations specifically): the chi square test operates over the full version distribution across ten countries, not just mean staleness across four, and detects a real, if modest, association the coarser comparison missed.

Geographic clusters and behavior. A Kruskal Wallis test was used to evaluate whether DBSCAN clusters differ in risk or anomaly score.

Metric	H	dof	p	ϵ^2	Effect size	n clusters	n routers
Risk score	2174.1	117	<0.001	0.090	moderate	118	23,041
Anomaly score	350.1	117	<0.001	0.010	small	118	23,041

Geographic cluster membership is significantly associated with risk score at a moderate effect size (which dense geographic group a router falls into meaningfully predicts its risk score, not just its location) and significant but small for anomaly score. §8 discusses why this gap between the two effect sizes is itself informative.

Metric redundancy. Of 28 pairwise correlations across the eight router level metrics computed by this pipeline, exactly one pair exceeds the $|r| \geq 0.7$ redundancy threshold: risk score and suspiciousness ($r = 0.964$, $n = 48,235$). This is expected rather than a new finding: suspiciousness’s underlying formula includes risk score directly as a weighted input (0.25 of its total), so the two are partially the same computation by construction, not independently corroborating signals. No other pair reaches the flagged threshold; the next highest, anomaly score vs. cluster distance ($r = -0.640$), and entropy vs. stability ($r = -0.598$), are moderate but distinct relationships worth noting rather than treating as redundant: routers far from their behavioral cluster centroid tend to have less negative (i.e., less anomalous by IsolationForest’s convention) scores, and routers with more regular observation timing (high stability) tend toward lower interval entropy, which is close to definitionally true given how both are derived from the same underlying observation interval data, rather than an independent empirical discovery.

8. Discussion

The results show that geography and network behavior are meaningfully connected. The chi square tests and the Kruskal Wallis results in Section 7.6 provide two independent lines of evidence for this conclusion. Floodfill capability, bandwidth tier, and protocol version all vary by country in ways that are statistically significant. Geographic clusters built only from spatial information also differ in risk score. This directly answers the study’s central empirical question. Geography is not just a visual overlay on the network. It is a structural dimension that shapes how routers behave. You can revisit this through geographic correlation.

Sections 7.3 and 7.6 refine the observation that Iran has many routers into a more specific picture. Iran’s floodfill rate is about one sixth of the US rate and is distinct from every other large population except Russia. This suggests a router population that leans toward newer or more resource constrained nodes rather than long running, high bandwidth infrastructure. At the same time, Iran’s IPv6 adoption is unusually high, reaching levels two to four times higher than several Western countries.

Combined with the structural findings in Section 7.3, which show heavy concentration in two state-linked ASNs and a single dense cluster centered on Tehran, the overall pattern is consistent with circumvention driven usage over state infrastructure. This cannot be proven from router data alone, but the combination of geographic, capability, and ASN patterns supports this interpretation. You can revisit this through Iran circumvention patterns.

Section 7.4 shows that the increase in Tehran's share of network activity is real and statistically supported. It also shows that the increase is reactive rather than anticipatory. The shift begins after the most intense strike days, not during them. Because the dataset covers only ten days and includes no pre-event baseline, the result must be interpreted cautiously.

The shift is presented as a suggestive pattern for future longitudinal work. It is not claimed as evidence of a causal relationship. Other explanations remain possible, including ISP reclassification or normal week-to-week variation. You can revisit this through Tehran activity analysis.

The Kruskal Wallis result for risk score should not be interpreted as geography causing risk score differences. The risk score formula includes country and ASN risk as direct weighted inputs. This means part of the association is built into the definition of the metric.

The anomaly score result is more informative. IsolationForest does not use geography as an input, yet geographic clusters still show a small but significant difference in anomaly score. This suggests that some geographic signal persists even when the metric is not designed to capture it. You can revisit this through IsolationForest behavior.

The implementation identification result as a methodological contribution

Section 6.1's negative result is important. It shows that passive netDb observation cannot be used to identify client implementations. This is a useful contribution for future I2P measurement work. Any project that wants to classify routers by implementation will need a different approach, such as active probing or a new data source. You can revisit this through I2P implementation limits.

9. Limitations

- Two vantage points do not provide a full census of the I2P network. Section 7.5 shows that the Java I2P and i2pd routers overlap on only about thirty percent of the routers they observe. This means each vantage point sees a different slice of the network. A study with more vantage points, especially ones placed in different network or geographic positions, would likely reveal routers that neither implementation observed. You can revisit this through vantage point coverage.
- Geolocation accuracy is another limitation. MaxMind's GeoLite2 databases provide estimates, especially at city level precision. Accuracy varies by region and ISP. This affects any result that relies on geographic coordinates or city level clustering.
- Implementation identification is not possible from netDb data, as explained in Section 6.1. Any future work that wants to classify routers by client implementation will need a different measurement approach. You can revisit this through I2P implementation limits.
- DBSCAN cluster counts depend on the chosen radius. Section 7.2 shows that cluster boundaries shift when the radius changes. The Adjusted Rand Index values between 0.48 and 0.65 indicate moderate stability, but not enough to treat any specific cluster count as ground truth. You can revisit this through DBSCAN sensitivity.
- Risk and suspiciousness scores do not have historical records. The pipeline recomputes these metrics fresh on every run and does not store timestamped versions. Because of this, the dataset cannot support claims about how a country's risk or suspiciousness changed over time. You can revisit this through risk scoring design.
- The ten day window is too short to characterize normal variance for the Tehran activity shift described in Section 7.4. With only one event and no baseline period, the study cannot determine whether similar shifts occur regularly or whether this one is unusual. You can revisit this through Tehran activity analysis

10. Future Work

- Historicizing risk and suspiciousness scoring is the most direct next step. Right now, the pipeline recomputes these metrics fresh on every run and does not store timestamped versions. A timestamped log table would allow researchers to test whether a country's aggregate risk or suspiciousness profile shifts around real world events. Section 7.3 shows why this matters. Without historical records, it is impossible to evaluate whether the patterns observed in this study represent short term noise or meaningful changes. You can revisit this idea through risk scoring design.
- Adding more vantage points is another clear direction. Section 7.5 shows that the two vantage points used here overlap on only about thirty percent of the routers they observe. Additional vantage points, especially ones placed in different network or geographic positions, would reduce this coverage gap. They would also help determine whether certain regions are more visible to specific implementations. You can revisit this through vantage point coverage.
- The anomaly detection, ASN reputation scoring, risk scoring, and dashboard infrastructure built alongside this study were not evaluated in this report. These components include an IsolationForest anomaly detector, PCA-based ASN reputation tiers, a composite router risk score, and behavioral K-means clustering. They were excluded because they address a different question. Their focus is on identifying routers that look unusual or risky, not on understanding geographic or version patterns. Even though they fall outside the scope of this report, they represent a natural extension for future security focused analysis of the same dataset. You can revisit this through I2P anomaly detection.

11. Conclusion

This study conducted a two vantage point census of 48,235 I2P routers across 161 countries and 2,866 Autonomous Systems to answer three questions about how the network is structured globally, and found affirmative, statistically supported answers to all three (§8): geography is a genuine structural axis of network behavior, not a cosmetic overlay; Iran's I2P presence is structurally distinct from other large populations in ways consistent with (though not proof of) circumvention driven usage over state infrastructure; and the timing of the Tehran activity share shift, while real, is reactive rather than anticipatory and should be read as suggestive rather than causal given this study's short, single event window. A fourth, originally central question (which client implementation is most prevalent) could not be answered from passive netDb data, and is reported here as a methodological finding in its own right (§6.1) rather than a gap. §4.5 documents, without hedging, where this project's completed work matched, exceeded, or diverged from its original proposal, including infrastructure built beyond the proposal's stated scope and deliberately excluded from these results. Together, §6 through §9 constitute this study's substantive contribution; this section does not restate their figures.

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